

1. Use an interpreter pattern to implement the addition and subtraction rules of two complex numbers.

AdditionExpression

```
package experiment23_1;

class AdditionExpression implements Expression{

    private Expression leftExpression;

    private Expression rightExpression;

    public AdditionExpression(Expression leftExpression,
Expression rightExpression){

        this.leftExpression = leftExpression;

        this.rightExpression = rightExpression;

    }

    @Override

    public ComplexNumber interpret() {

        return

leftExpression.interpret().add(rightExpression.interpret());

    }

}
```

ComplexConstant

```
package experiment23_1;

class ComplexConstant implements Expression{
```

```

    private ComplexNumber value;

    public ComplexConstant(double real, double imaginary){
        value = new ComplexNumber(real, imaginary);
    }

    @Override
    public ComplexNumber interpret() {
        return value;
    }
}

```

ComplexNumber

```

package experiment23_1;

class ComplexNumber {
    private double real;
    private double imaginary;

    public ComplexNumber(double real, double imaginary) {
        this.real = real;
        this.imaginary = imaginary;
    }
}

```

```

    public ComplexNumber add(ComplexNumber other) {
        return new ComplexNumber(this.real + other.real,
this.imaginary + other.imaginary);
    }

    public ComplexNumber subtract(ComplexNumber other) {
        return new ComplexNumber(this.real - other.real,
this.imaginary - other.imaginary);
    }

    @Override
    public String toString() {
        return String.format("%f + %fi", real, imaginary);
    }
}

```

Expression

```

package experiment23_1;

interface Expression {
    ComplexNumber interpret();
}

```

SubtractionExpression

```

package experiment23_1;

class SubtractionExpression implements Expression{
    private Expression leftExpression;
    private Expression rightExpression;
    public SubtractionExpression(Expression leftExpression,
Expression rightExpression){
        this.leftExpression = leftExpression;
        this.rightExpression = rightExpression;
    }

    @Override
    public ComplexNumber interpret() {
        return
leftExpression.interpret().subtract(rightExpression.interpret())
;
    }
}

```

Test

```

package experiment23_1;

public class Test {

```

```

    public static void main(String[] args) {
        Expression complexConstant1 = new
ComplexConstant(3,2);

        Expression complexConstant2 = new
ComplexConstant(1,7);

        // 3 + 2i + 1 + 7i

        Expression addition = new
AdditionExpression(complexConstant1, complexConstant2);

        System.out.println("Addition Result: " +
addition.interpret());

        // 3 + 2i - (1 + 7i)

        Expression subtraction = new
SubtractionExpression(complexConstant1,
complexConstant2);

        System.out.println("Subtraction Result: " +
subtraction.interpret());
    }
}

```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\Intelli
```

```
Addition Result: 4.000000 + 9.000000i
```

```
Subtraction Result: 2.000000 + -5.000000i
```

```
Process finished with exit code 0
```